

Idea de las Máquinas de Boltzmann

Plan de Ataque

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Qué vamos a aprender en esta sección

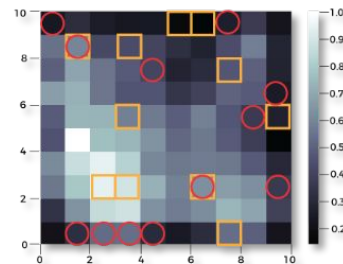
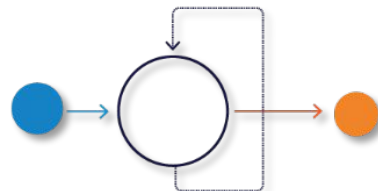
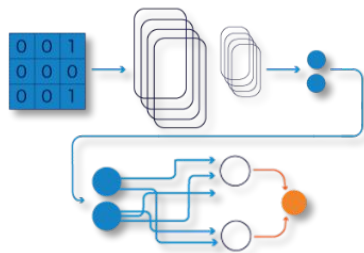
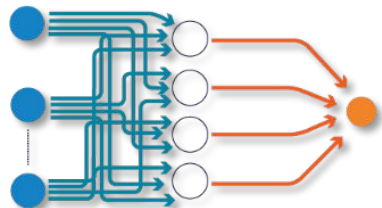
- La Máquina de Boltzmann
- Modelos basados en Energía (EBM)
- Máquinas de Boltzmann Restringidas (RBM)
- Divergencia Contrastante (CD)
- Red de Creencia Profunda (DBN)
- Máquinas de Boltzmann Profundas (DBM)

Máquinas de Boltzmann

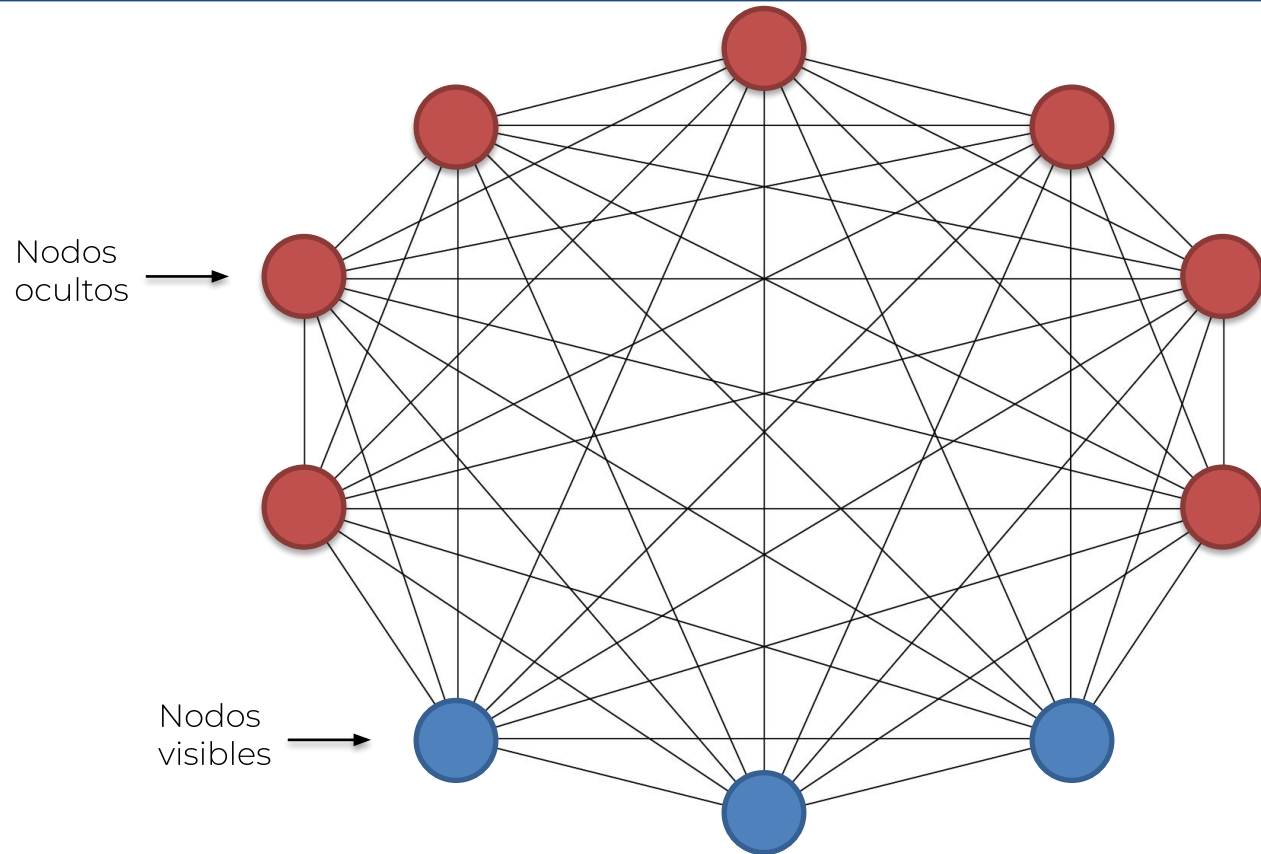
Máquinas de Boltzmann

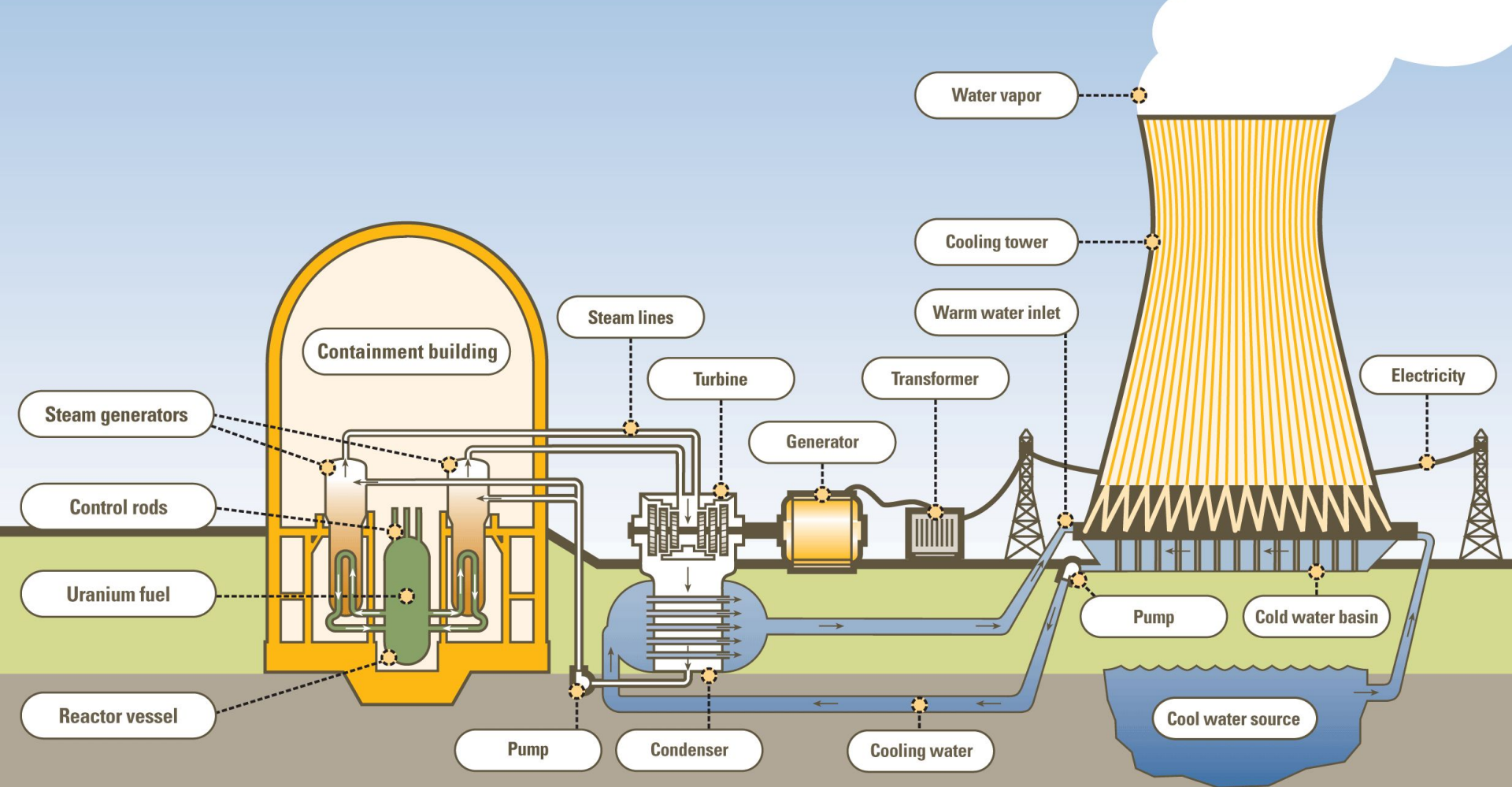
Supervisado	Redes neuronales artificiales	Utilizado para regresión y clasificación
	Redes neuronales convolucionales	Utilizado para la visión por ordenador
	Redes neuronales recurrentes	Utilizado para el análisis de series temporales
No Supervisado	Mapas autoorganizados	Utilizado para la detección de características
	Máquinas de Boltzmann Profundas	Utilizado para los sistemas de recomendación
	AutoEncoders	Utilizado para los sistemas de recomendación

Máquinas de Boltzmann

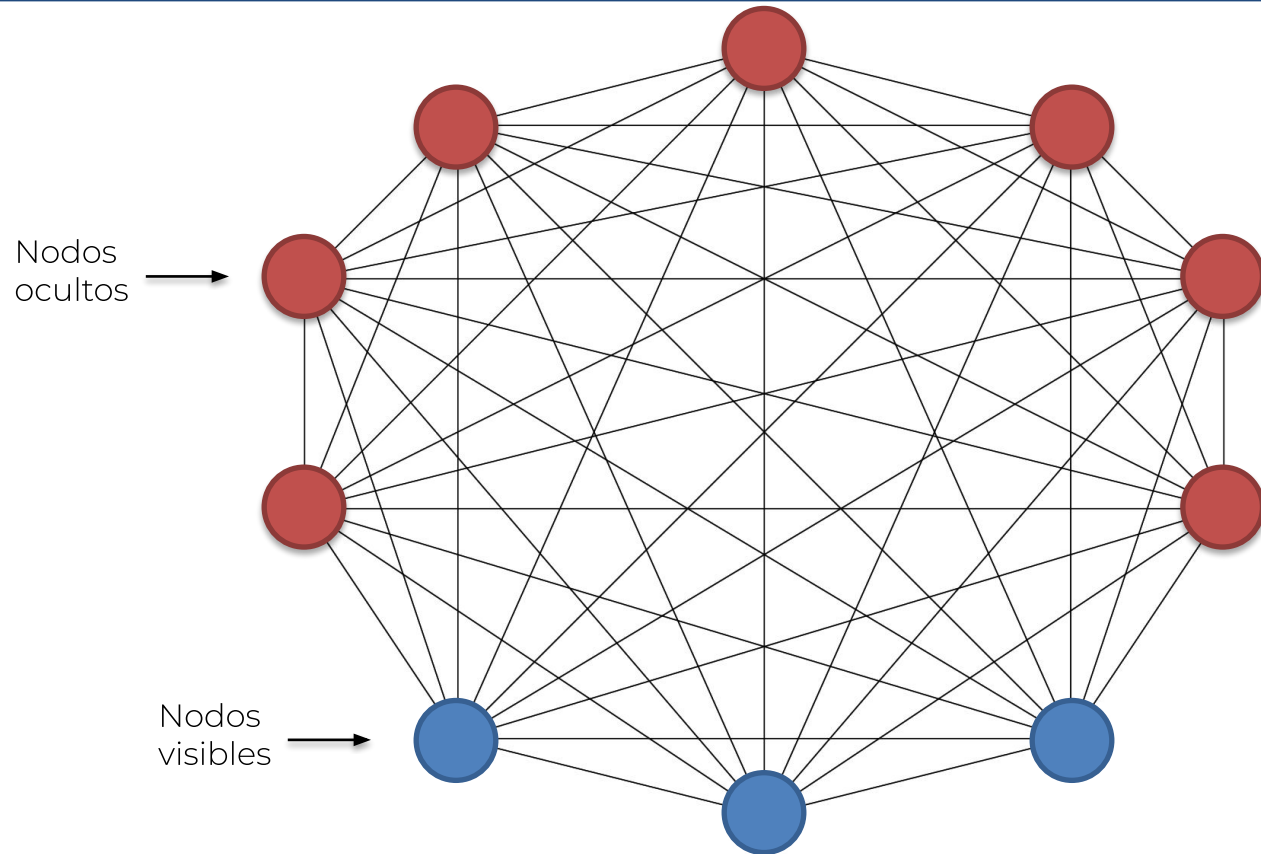


Máquinas de Boltzmann





Máquinas de Boltzmann

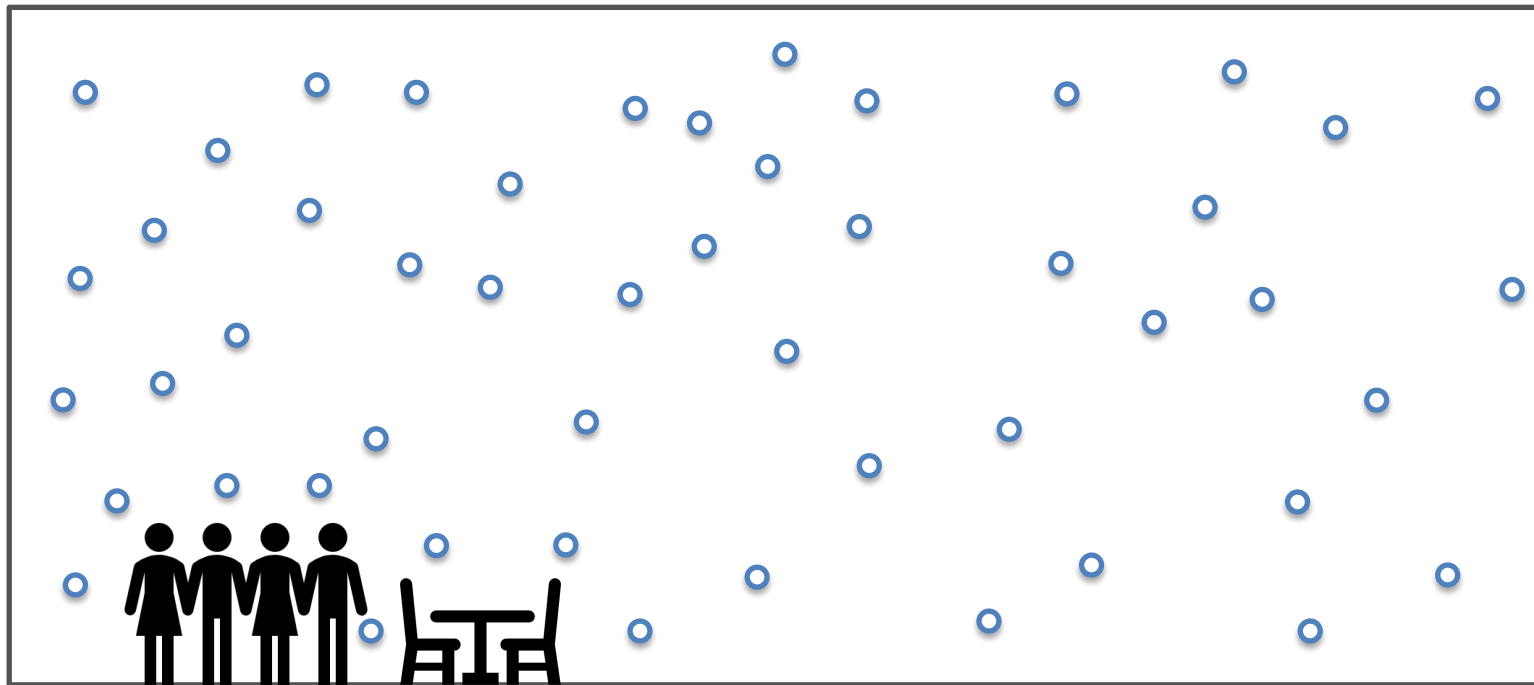


Modelos basados en Energía

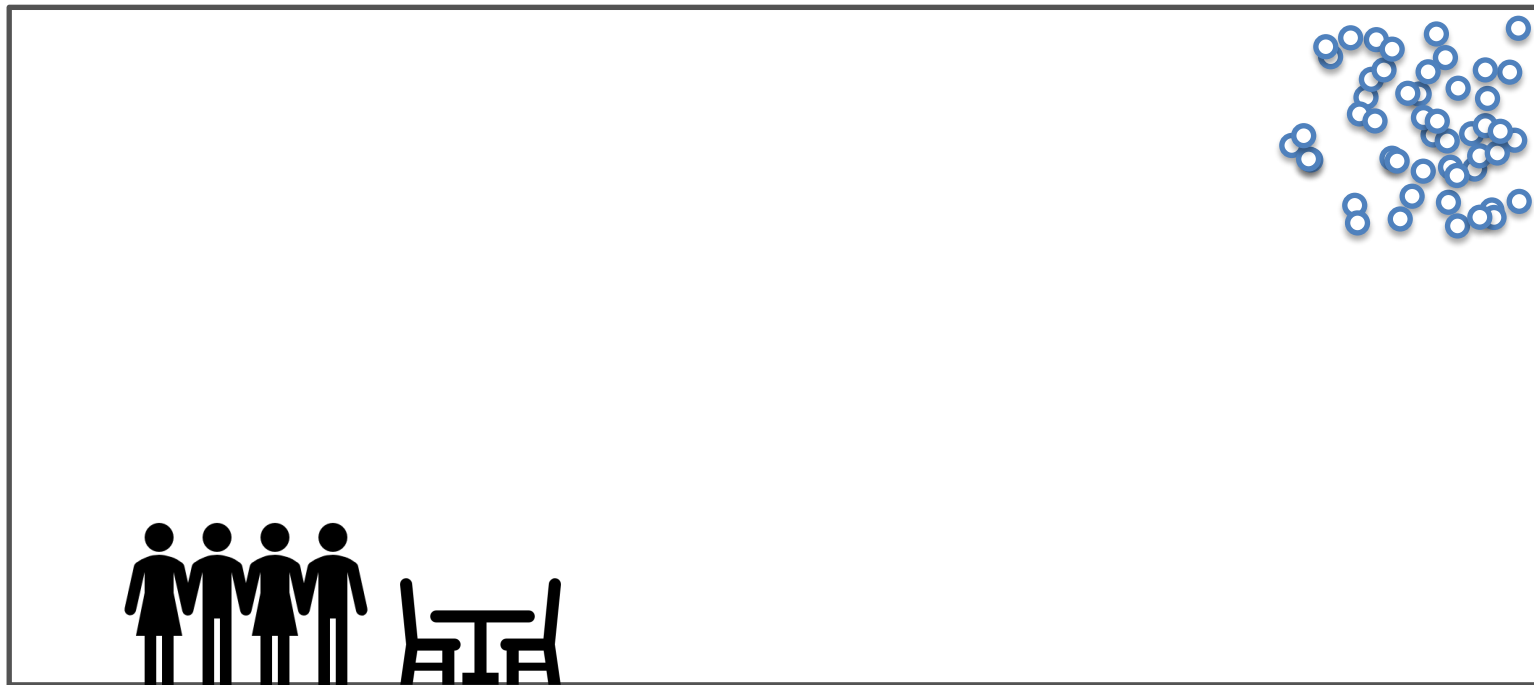
Modelos basados en Energía

$$p_i = \frac{e^{-\varepsilon_i/kT}}{\sum_{j=1}^M e^{-\varepsilon_j/kT}}$$

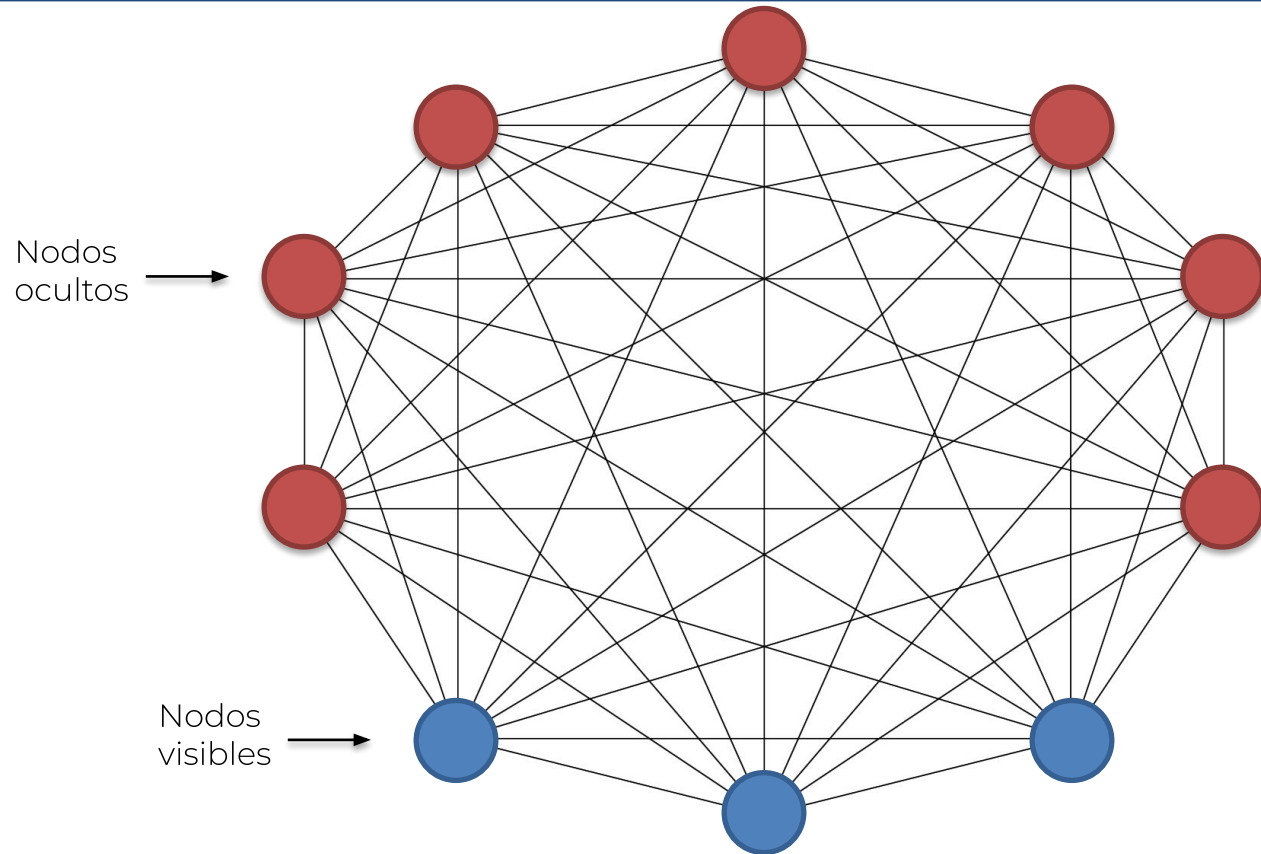
Modelos basados en Energía



Modelos basados en Energía



Máquinas de Boltzmann



Modelos basados en Energía

$$E(v, h) = - \sum_i a_i v_i - \sum_j b_j h_j - \sum_i \sum_j v_i w_{i,j} h_j$$

$$P(v, h) = \frac{1}{Z} e^{-E(v, h)}$$

Modelos basados en Energía

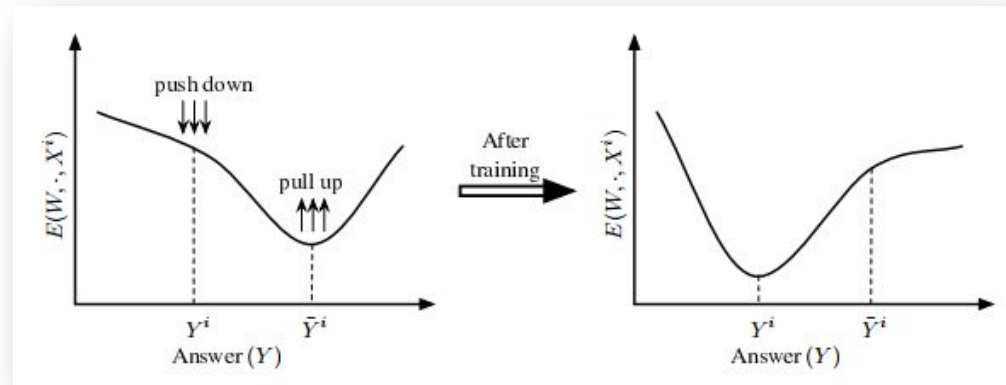
Lectura adicional:

A Tutorial on Energy-Based Learning

de Yann LeCun et al. (2006)

Link:

<http://yann.lecun.com/exdb/publis/pdf/lecun-06.pdf>



Modelos basados en En

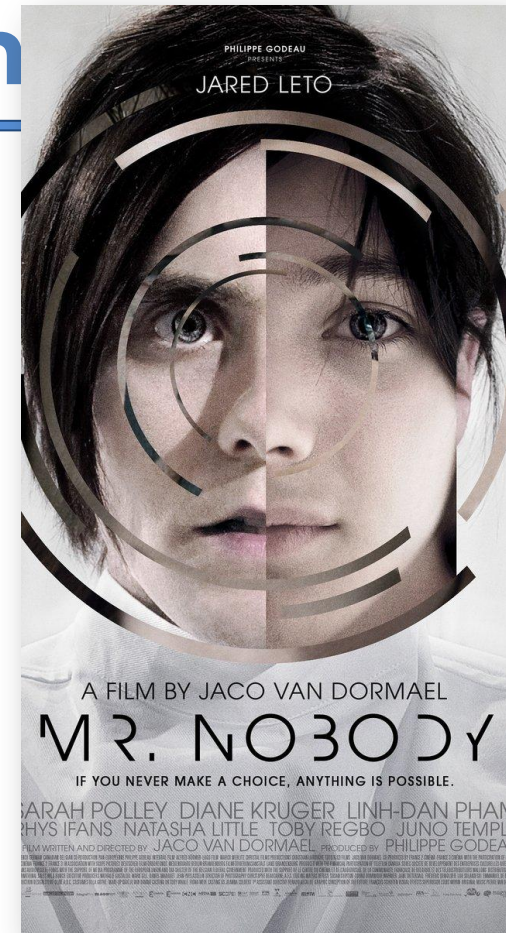
Lectura Adicional:

Mr. Nobody (Película)

de Jaco Van Dormael (2009)

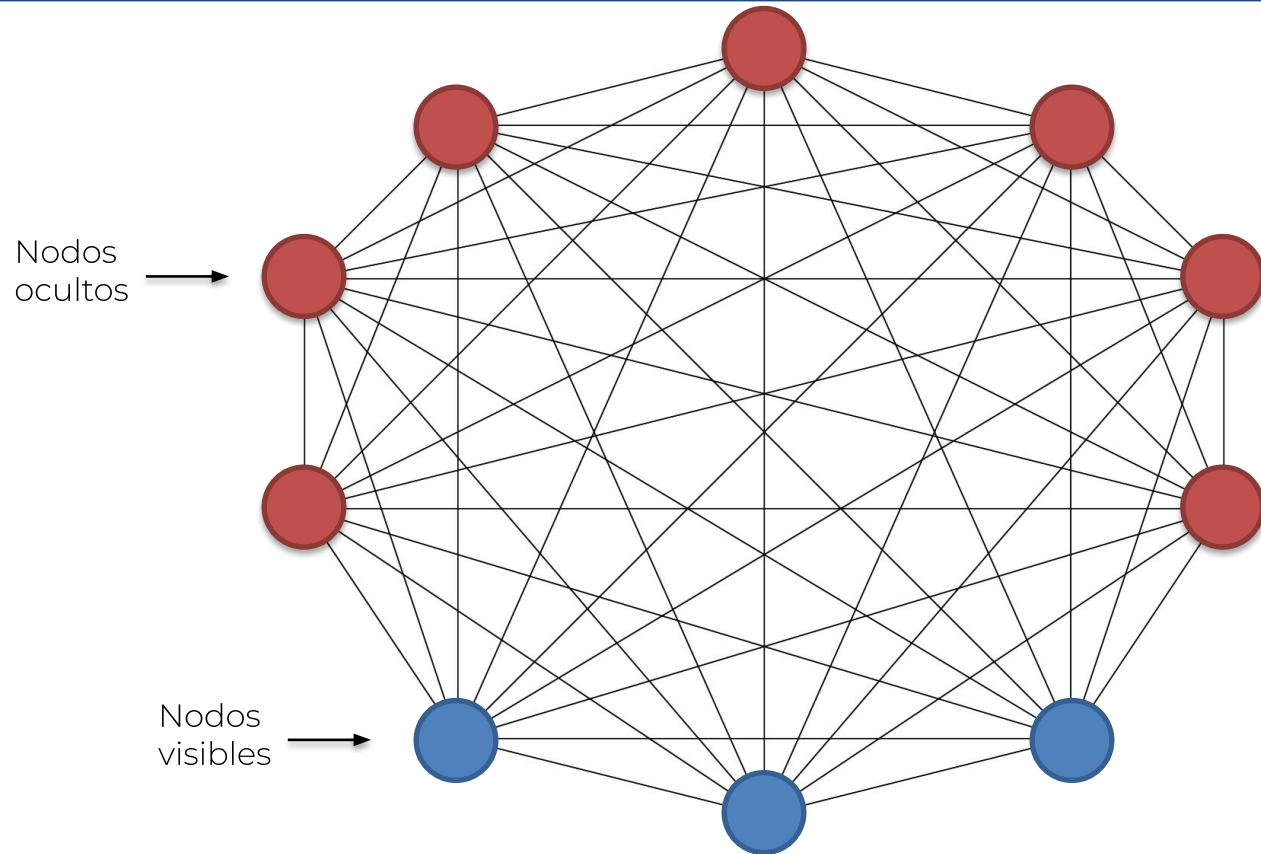
Link:

<http://www.imdb.com/title/tt0485947/>

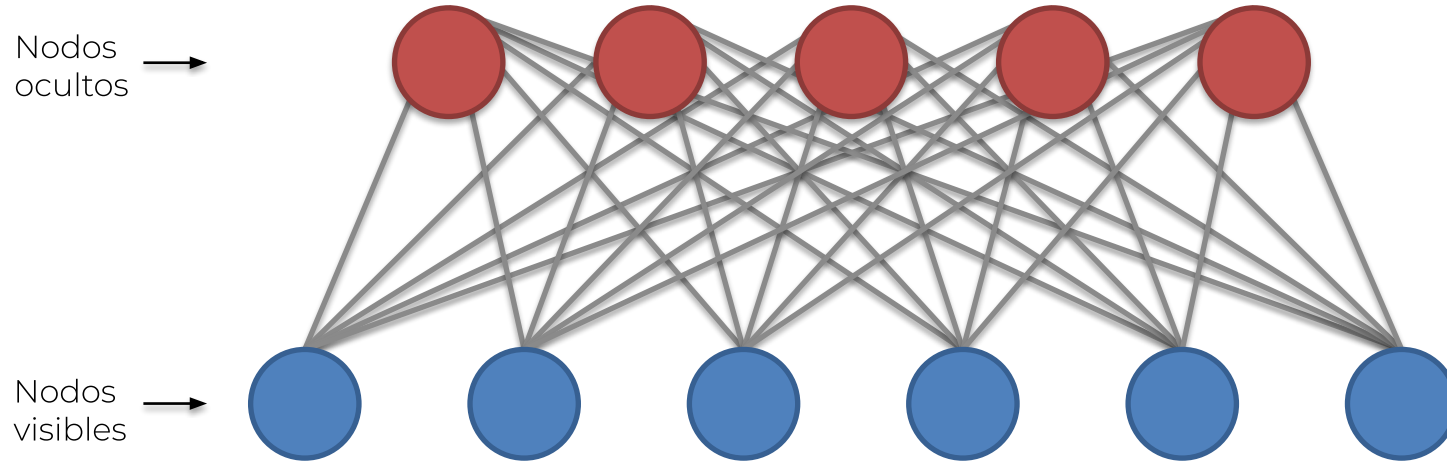


Máquinas de Boltzmann Restringidas (RBM)

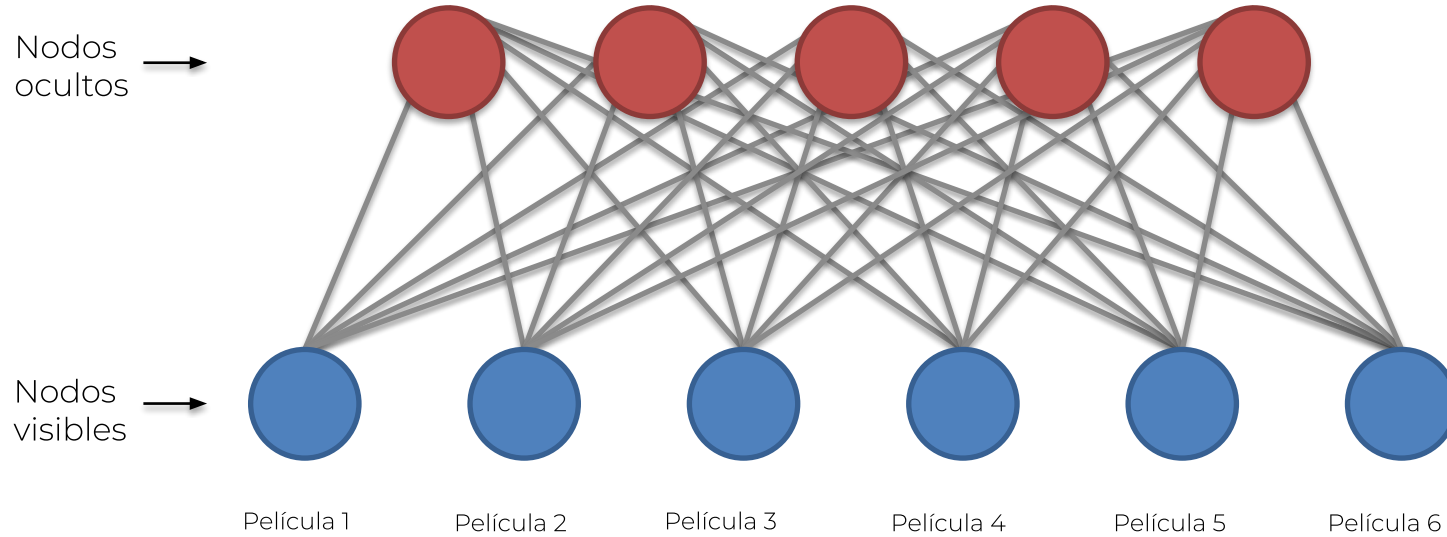
Máquinas de Boltzmann



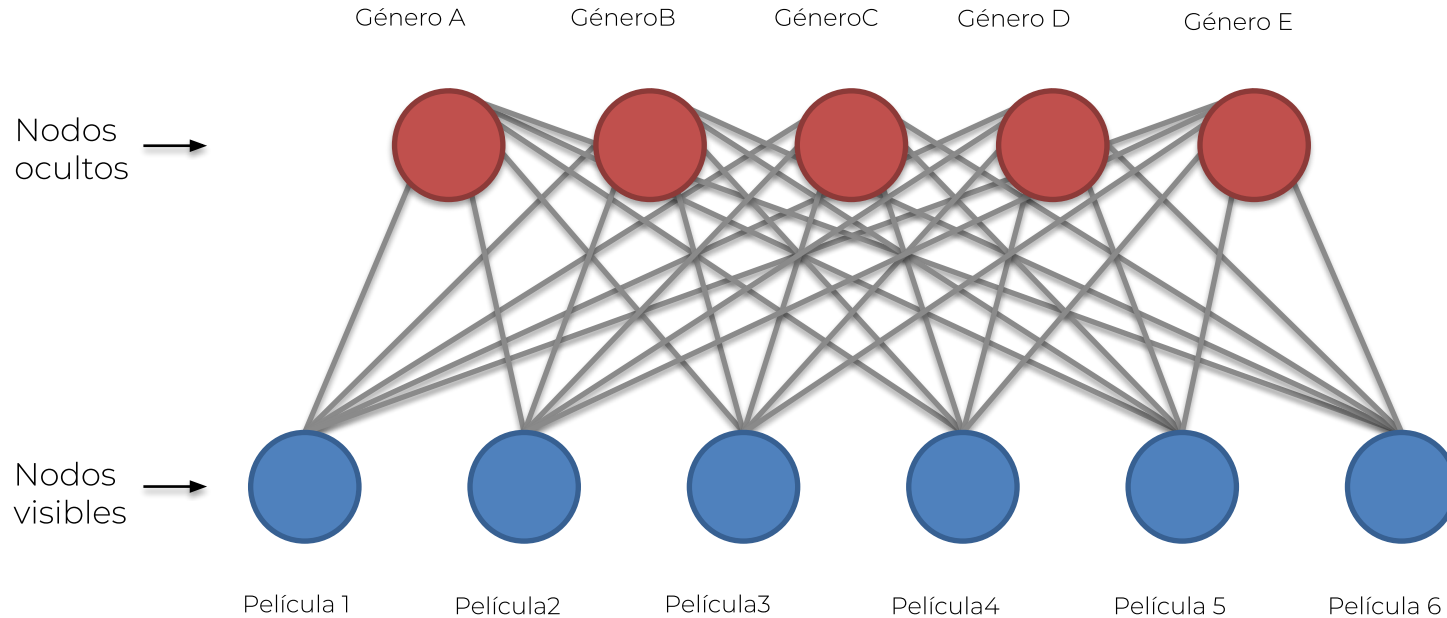
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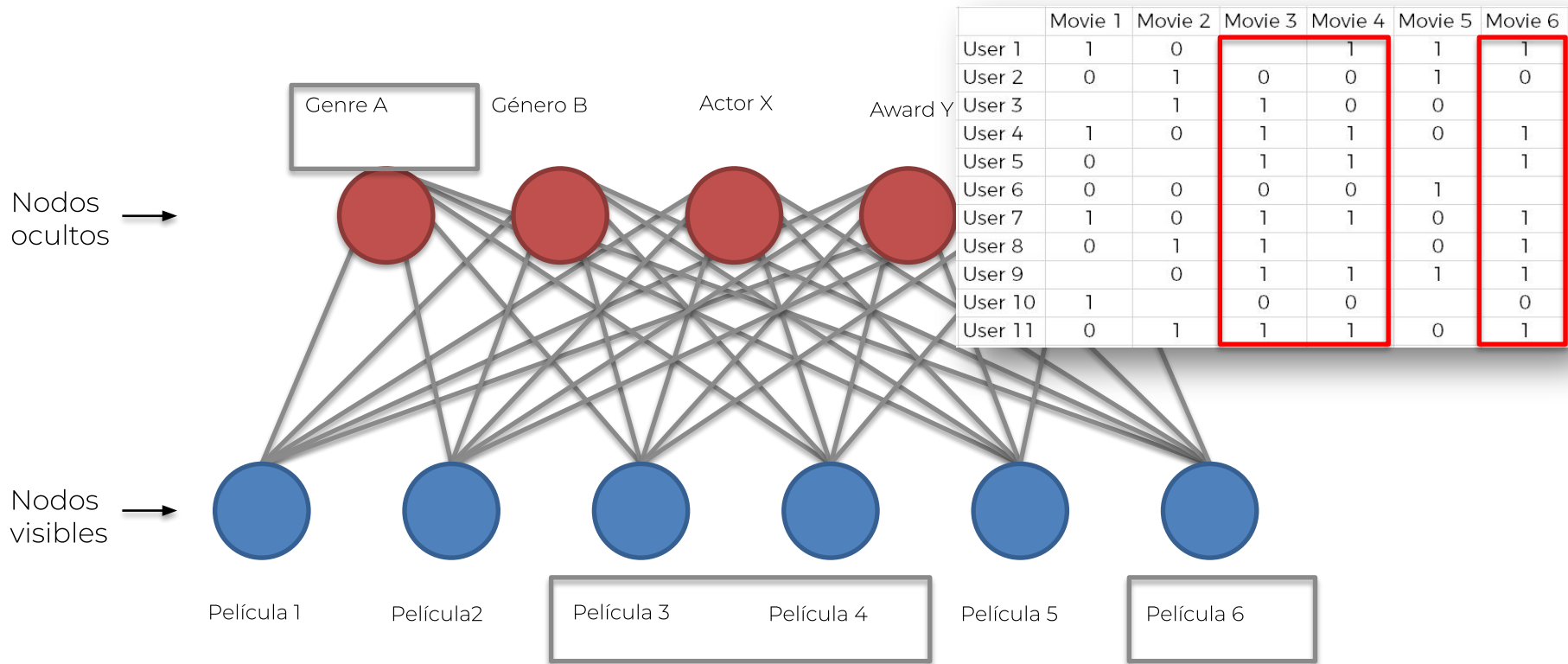
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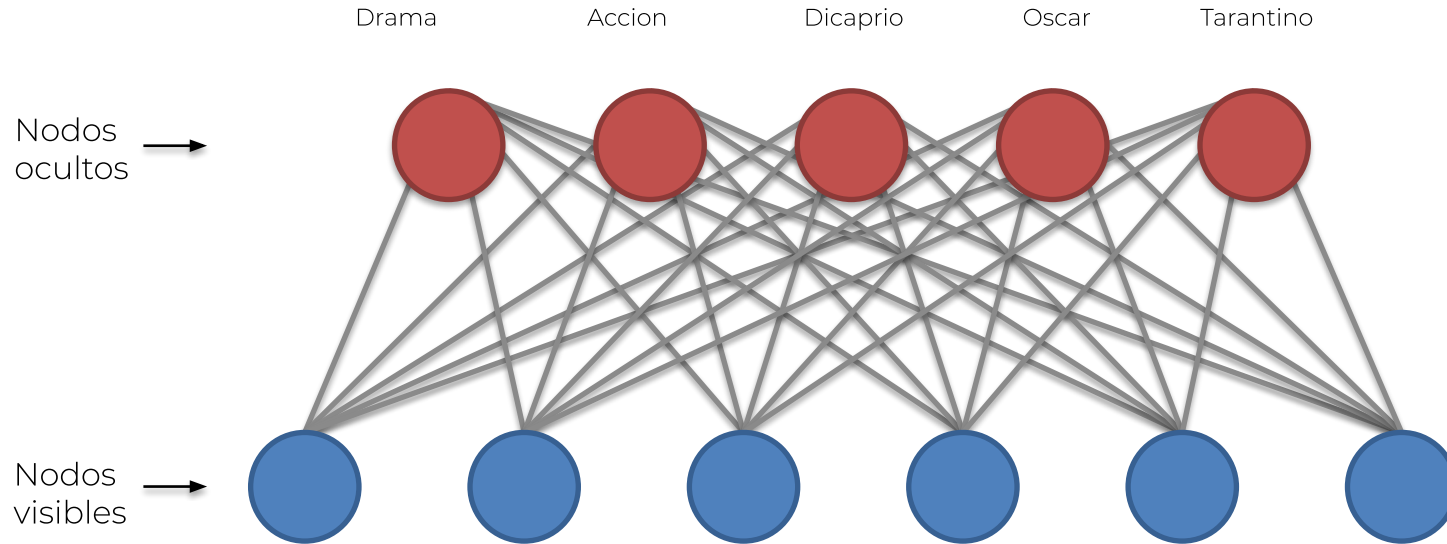
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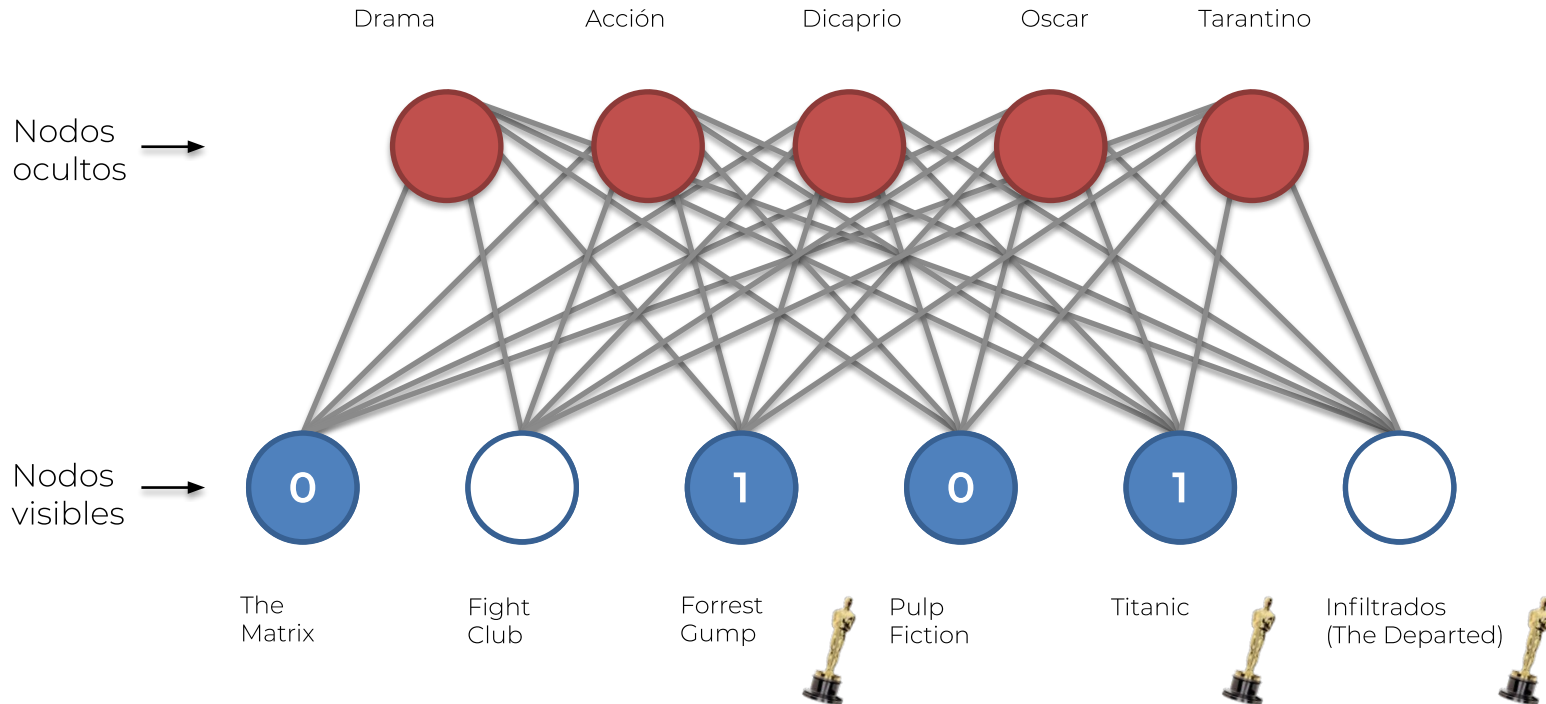
Máquinas de Boltzmann Restringidas (RBM)



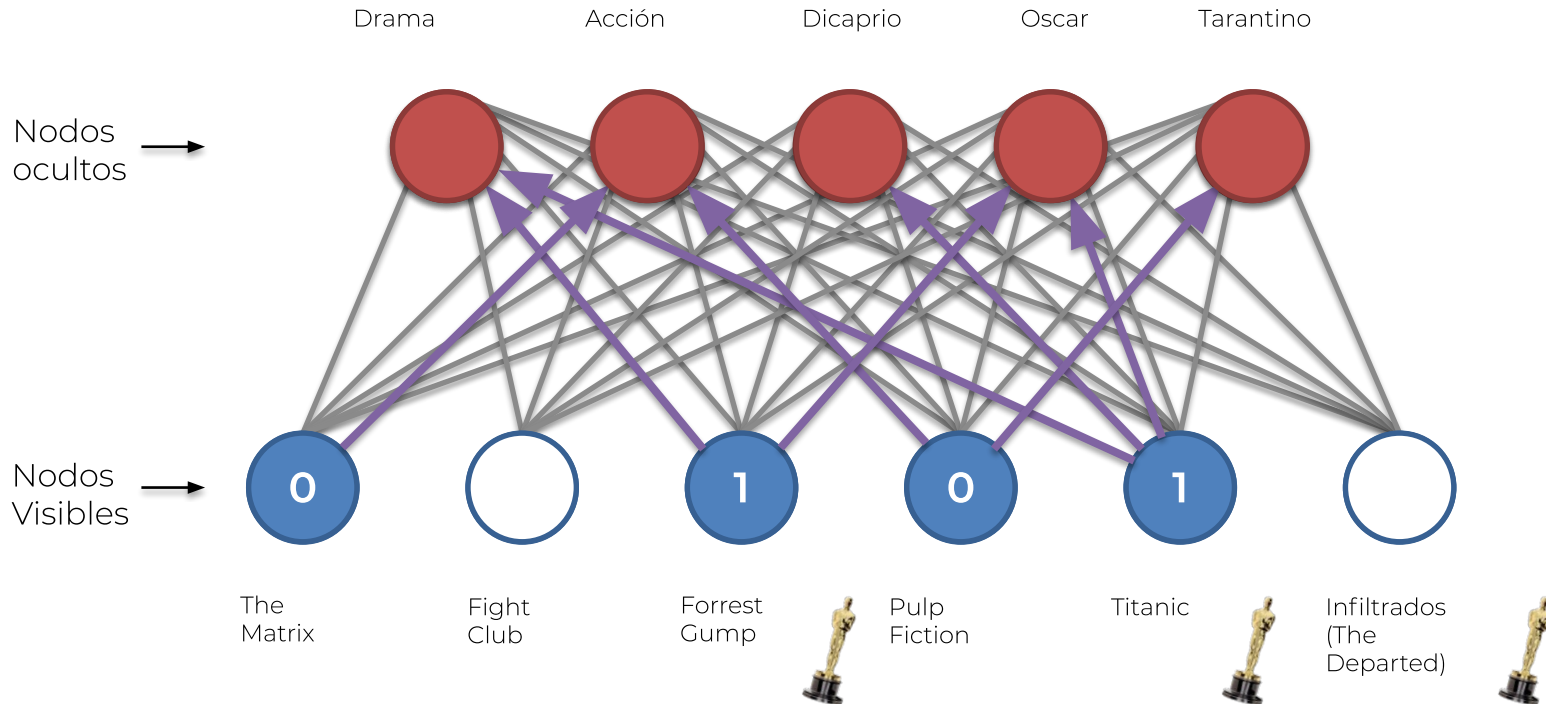
Máquinas de Boltzmann Restringidas (RBM)



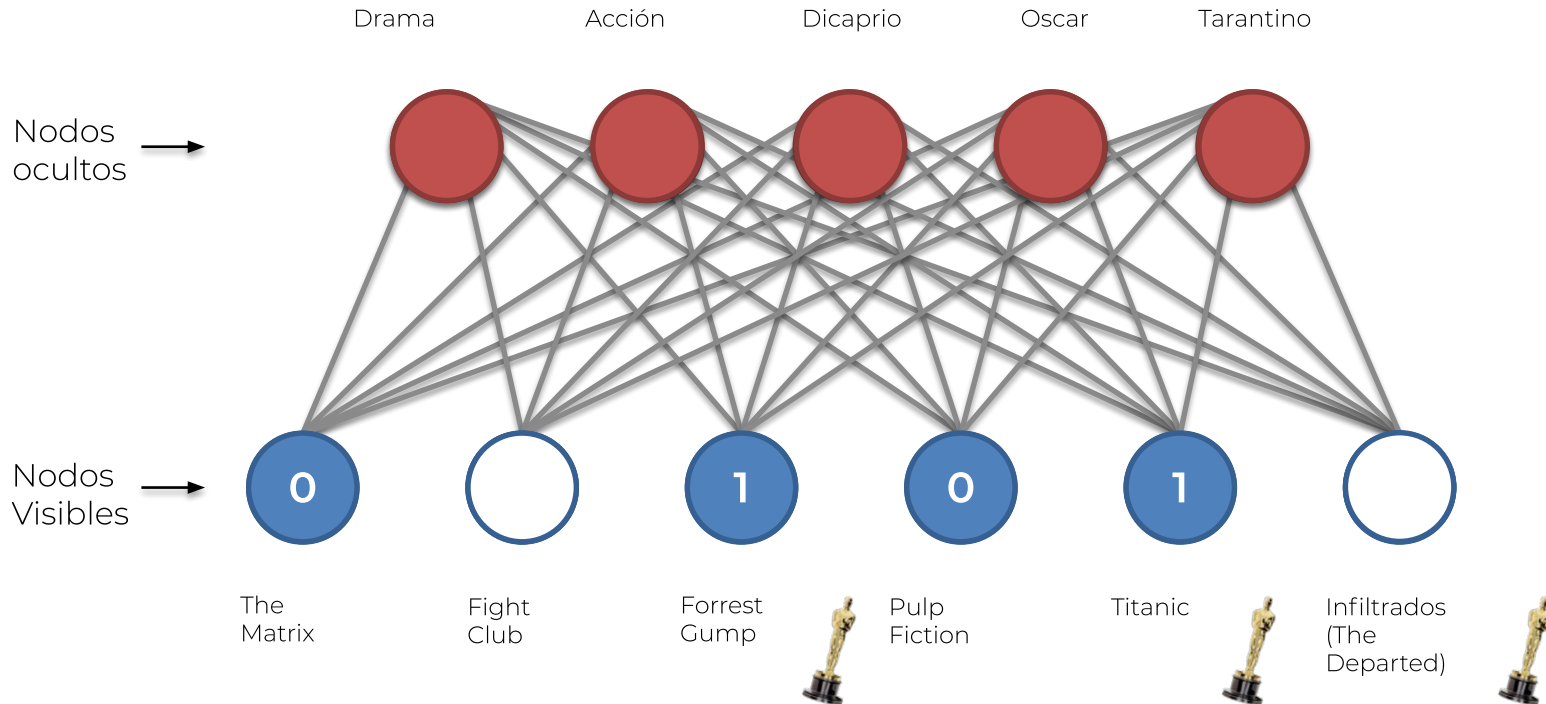
Máquinas de Boltzmann Restringidas (RBM)



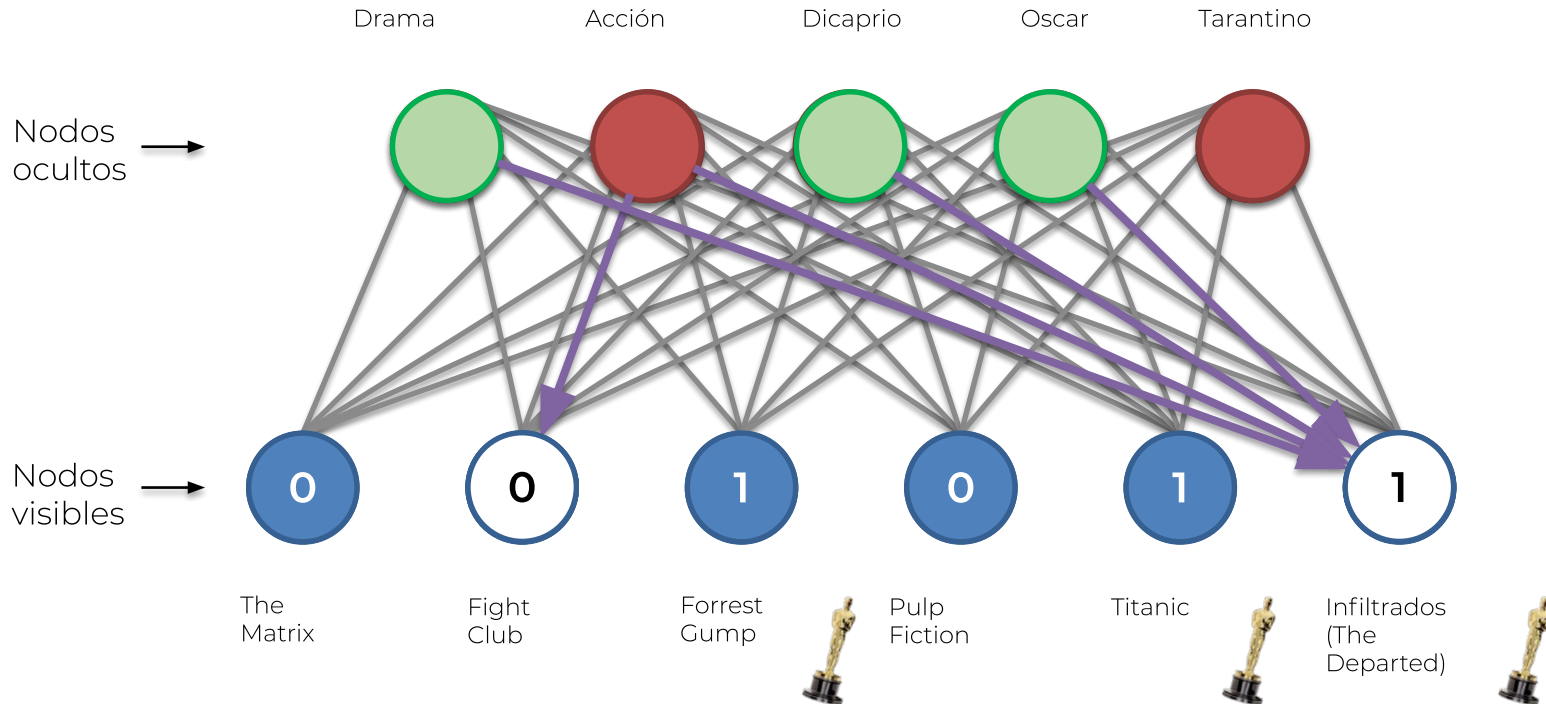
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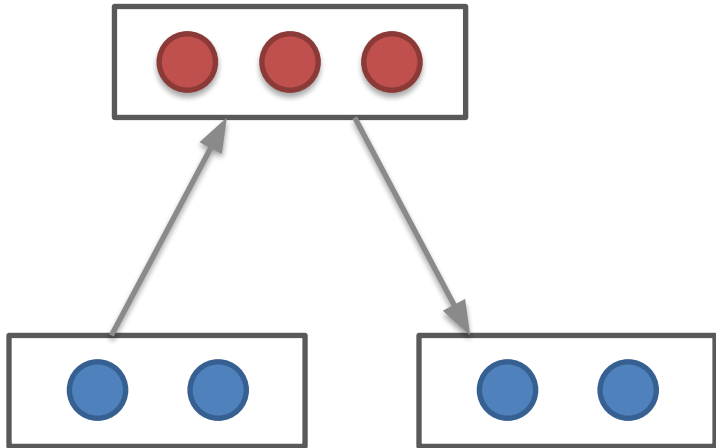


Máquinas de Boltzmann Restringidas (RBM)

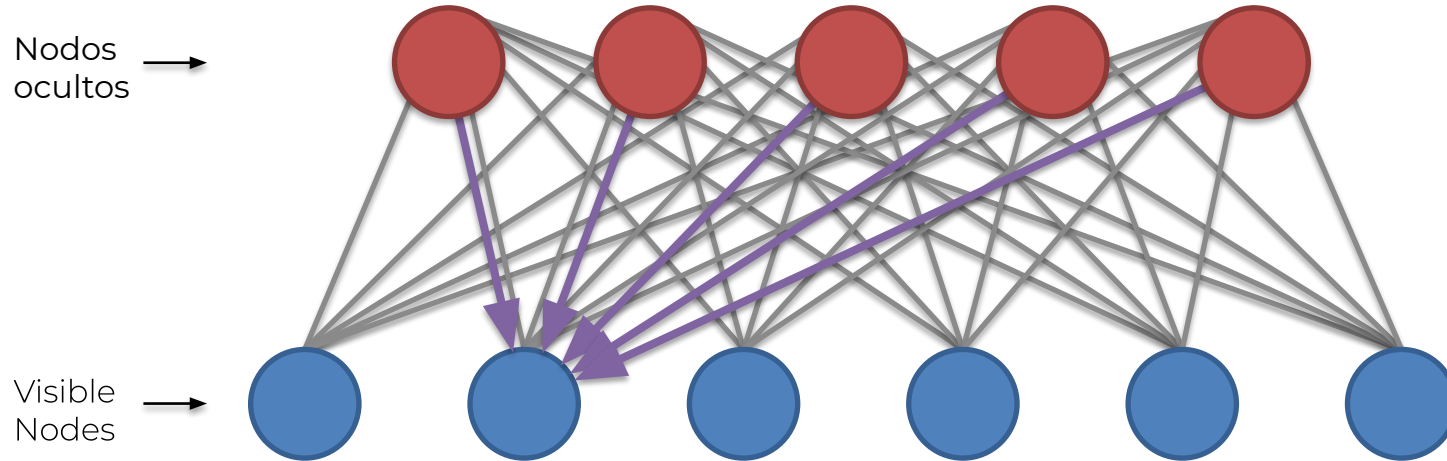


Divergencia contrastante

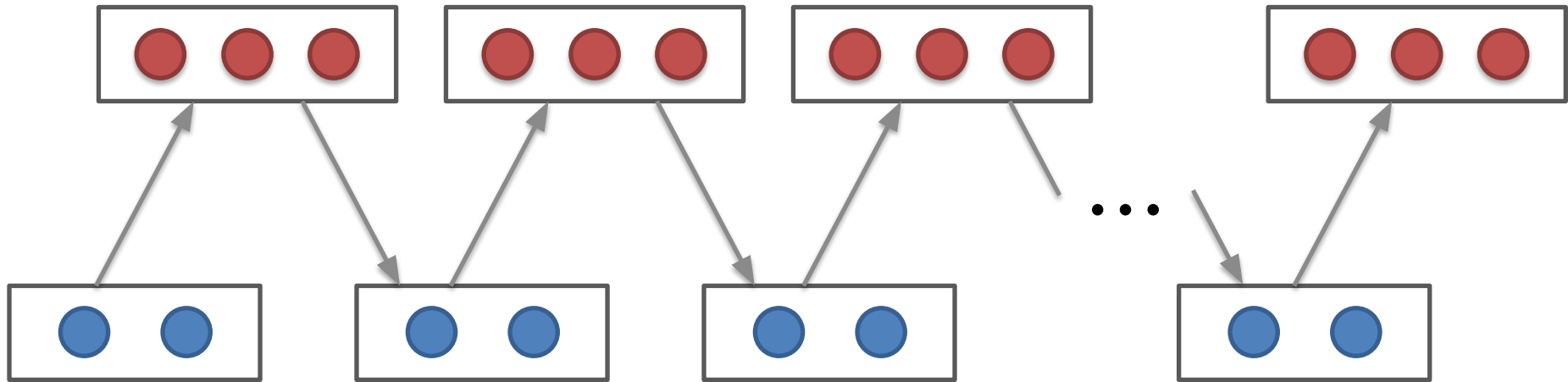
Divergencia Contrastante



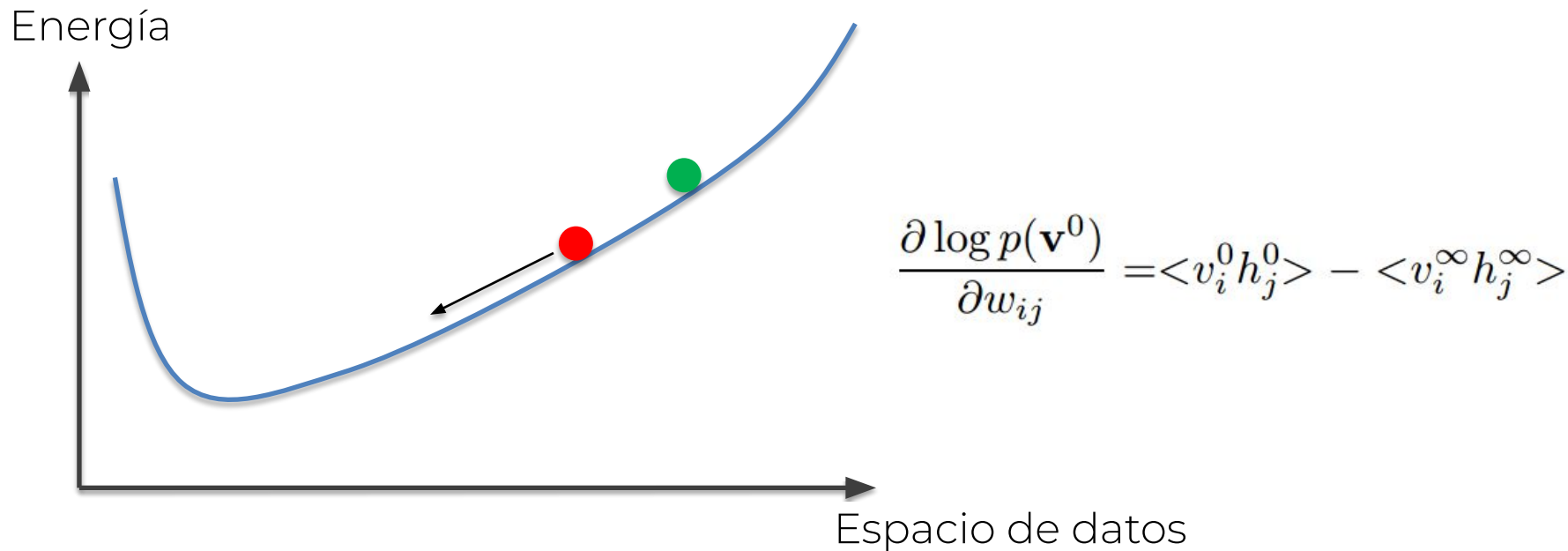
Máquinas de Boltzmann restringidas (RBM)



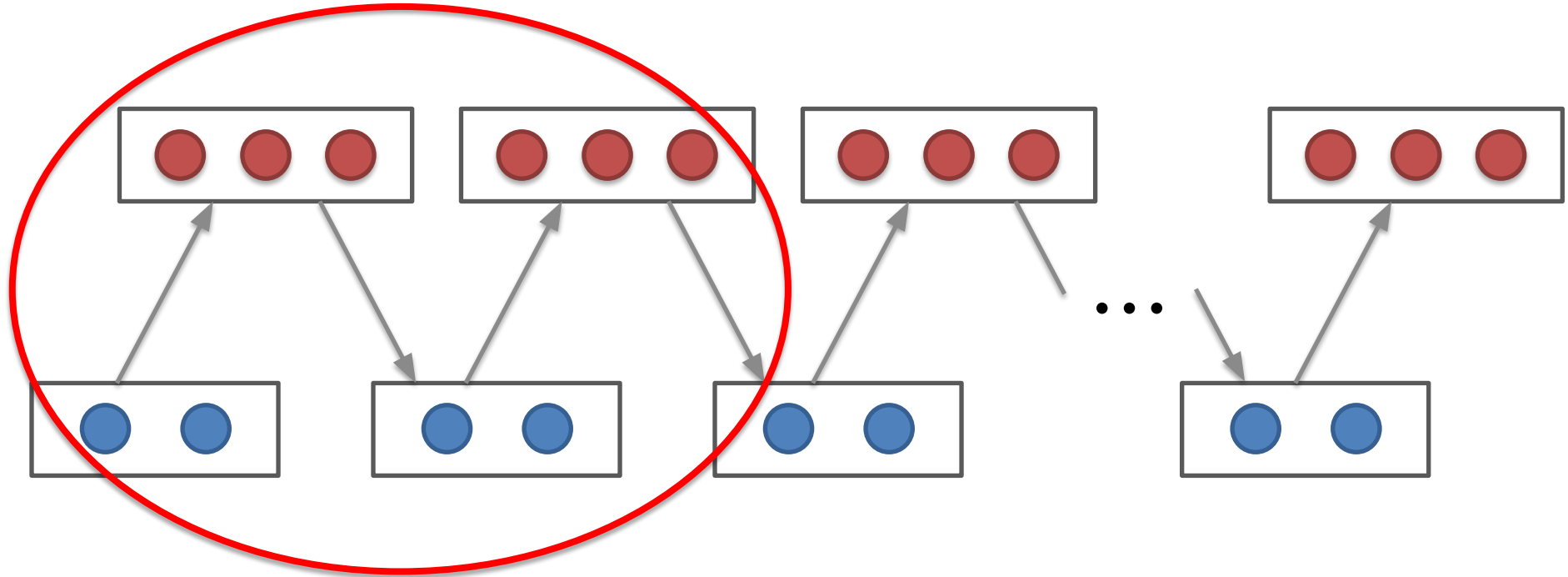
Divergencia Contrastante



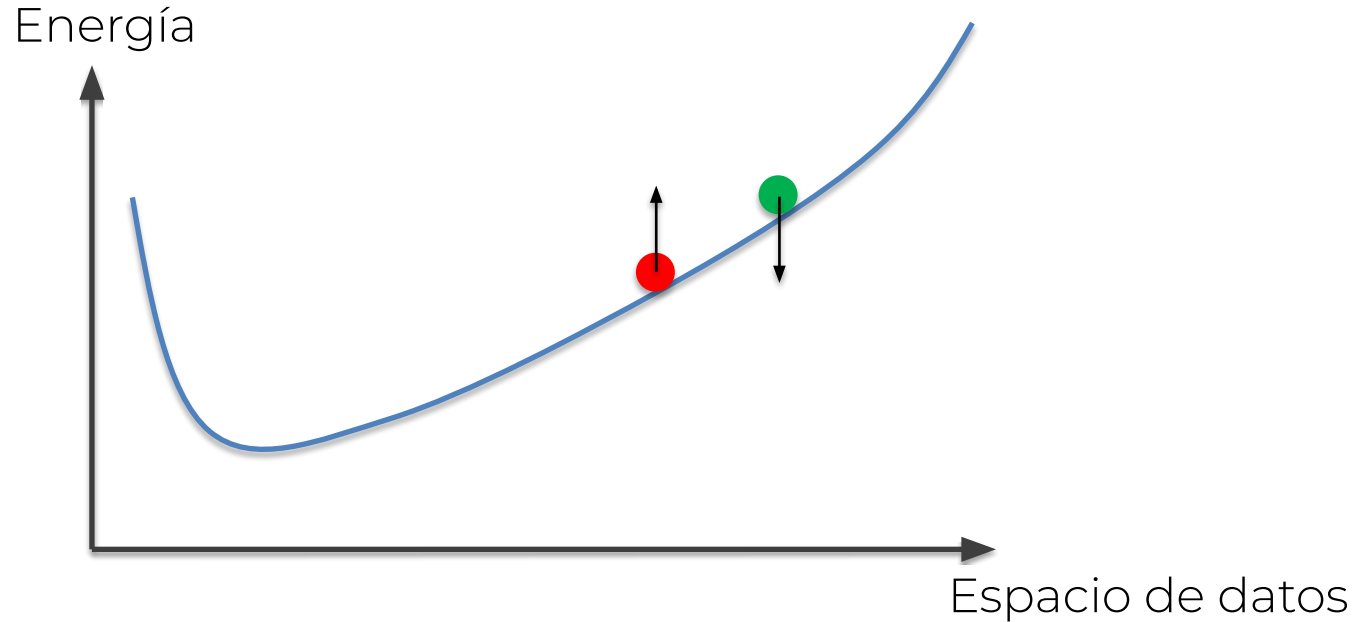
Divergencia Contrastante



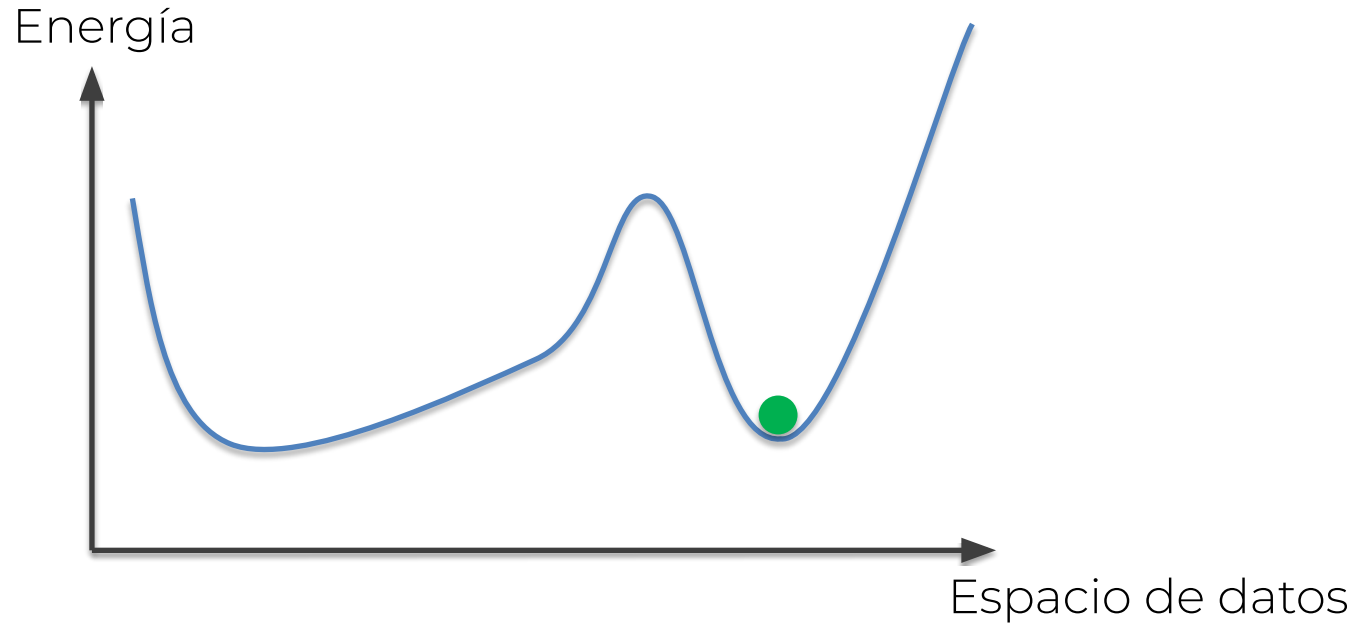
Divergencia Contrastante



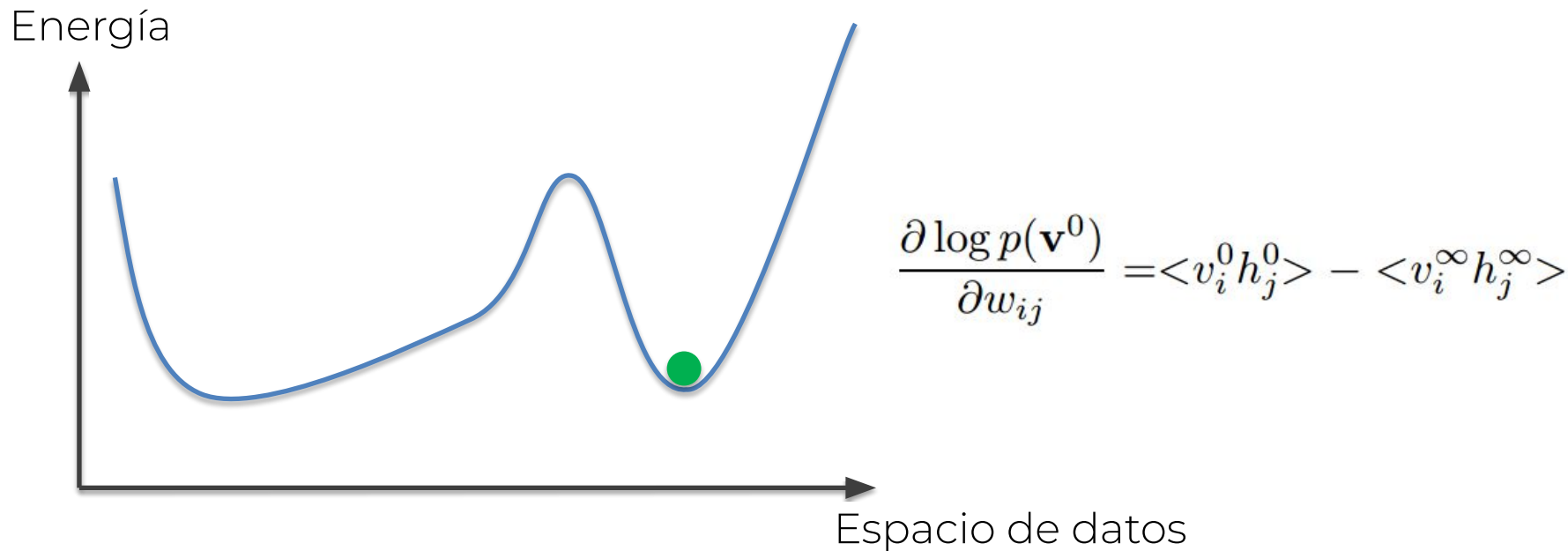
Divergencia Contrastante



Divergencia Contrastante



Divergencia Contrastante

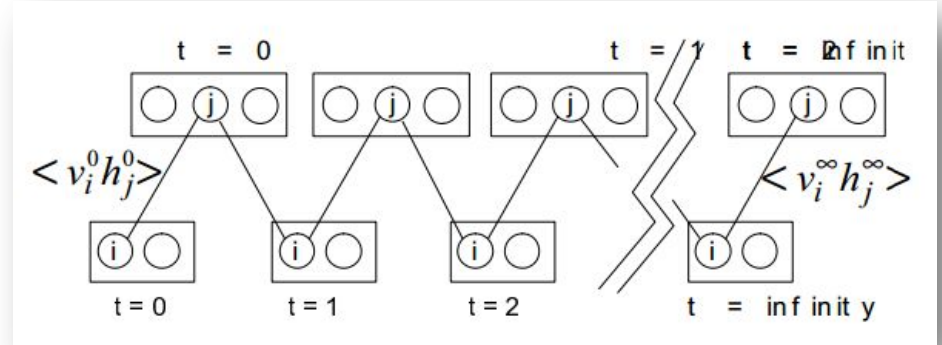


Divergencia Contrastante

Lectura adicional:

A fast learning algorithm for deep belief nets

de Geoffrey Hinton et al. (2006)



Link:

<https://www.cs.toronto.edu/~hinton/absps/fastnc.pdf>

Divergencia Contrastante

Additional Reading:

Notes on Contrastive Divergence

de Oliver Woodford (2012?)

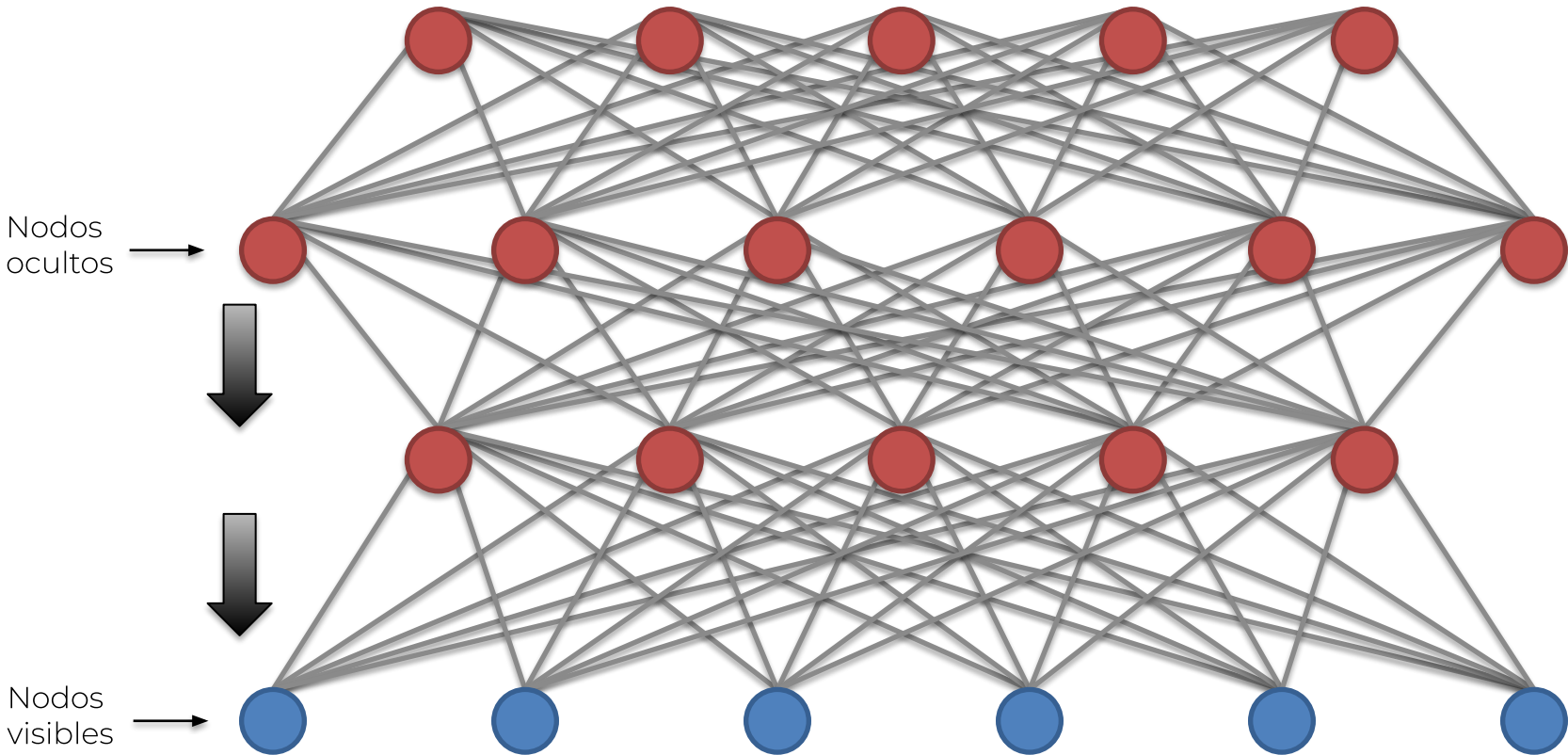
Link:

<http://www.robots.ox.ac.uk/~ojw/files/NotesOnCD.pdf>

$$\begin{aligned}\frac{\partial \log Z(\Theta)}{\partial \Theta} &= \frac{1}{Z(\Theta)} \frac{\partial Z(\Theta)}{\partial \Theta} \\ &= \frac{1}{Z(\Theta)} \frac{\partial}{\partial \Theta} \int f(x; \Theta) dx \\ &= \frac{1}{Z(\Theta)} \int \frac{\partial f(x; \Theta)}{\partial \Theta} dx \\ &= \frac{1}{Z(\Theta)} \int f(x; \Theta) \frac{\partial \log f(x; \Theta)}{\partial \Theta} dx \\ &= \int p(x; \Theta) \frac{\partial \log f(x; \Theta)}{\partial \Theta} dx \\ &= \left\langle \frac{\partial \log f(x; \Theta)}{\partial \Theta} \right\rangle_{p(x; \Theta)}\end{aligned}$$

Red de Creencia Profunda

Red de Creencia Profunda (DBN)



Red de Creencia Profunda (DBN)

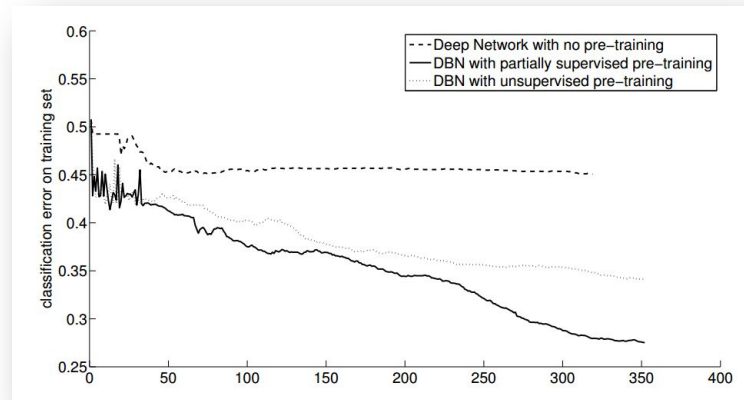
Lectura adicional:

Greedy Layer-Wise Training of Deep Networks

de Yoshua Bengio et al. (2006)

Link:

<http://www.iro.umontreal.ca/~lisa/pointeurs/BengioNips2006All.pdf>



Red de Creencia Profunda (DBN)

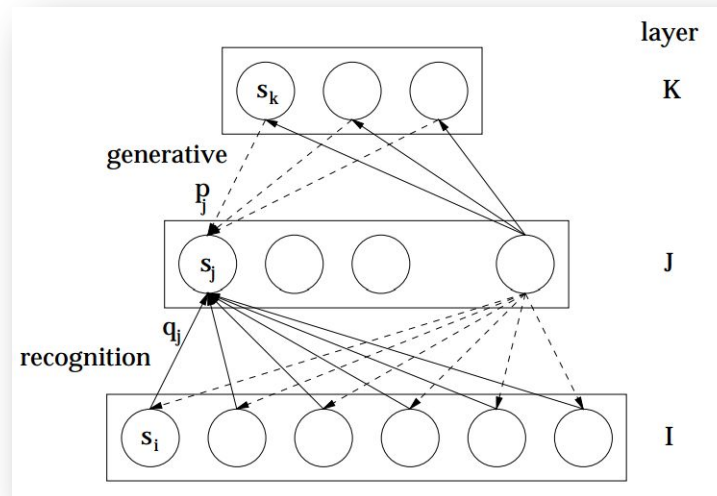
Lectura adicional:

The wake-sleep algorithm for unsupervised neural networks

de Geoffrey Hinton et al. (1995)

Link:

<http://www.gatsby.ucl.ac.uk/~dayan/papers/hdfn95.pdf>

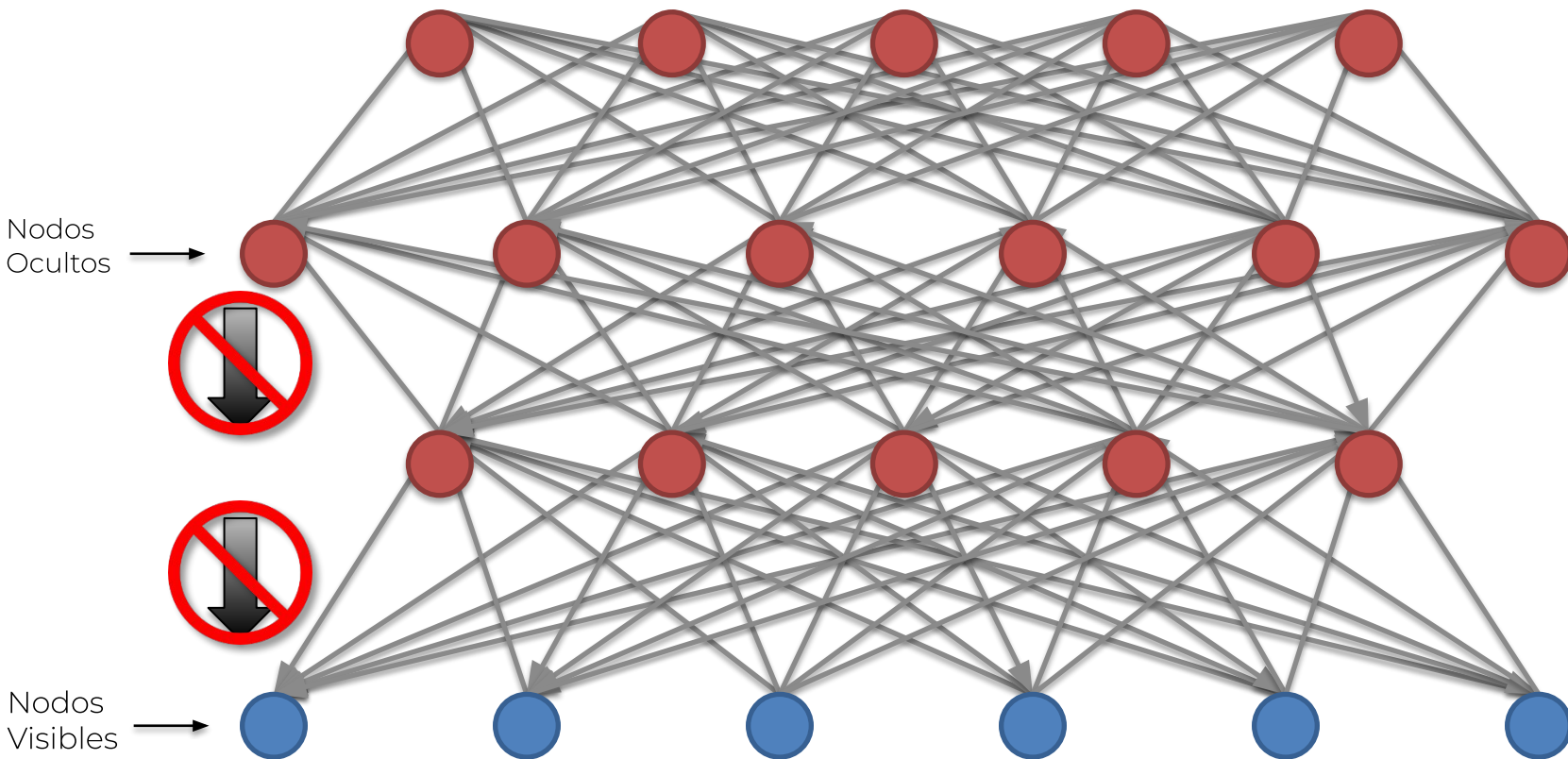


Máquinas de Boltzmann Profundas

Máquinas de Boltzmann Profundas (DBM)

DBN <> DBM

Máquinas de Boltzmann Profundas (DBM)



Máquinas de Boltzmann Profundas (DBM)

Additional Reading:

Deep Boltzmann Machines

de Ruslan Salakhutdinov et al. (2009?)

Link:

<http://www.utstat.toronto.edu/~rsalakhu/papers/dbm.pdf>

